General disclosure
I used ClaudeAI Sonnet 5 Medium Model to generate my code.
I consulted it September 27, 2026.

Code attribution
Prompt used to generate AI code:

Write a C++ program with bool extractIPv4(const std::string& str, unsigned long& outAddress, int& outPort) that finds one IPv4 address with an optional :port in a line of text.
Rules:
- Only digits, '.', and ':' can be part of a token; any other character is a separator. Split the line into maximal runs of those characters.
- A run must match the grammar in full (no extracting a valid piece from inside a longer run). Return the first run that does.
- Address: exactly 4 octets, 1-3 digits each, 0-255, no leading zeros (except "0").
- Optional port right after the 4th octet: 1-5 digits, 0-65535, no leading zeros (except "0"). If a colon is present and the port is invalid, reject the whole run.
- Reject empty octets, extra colons, and stray leading/trailing '.' or ':'.
- On failure: outAddress = 0, outPort = -1. On success outPort is -1 if there is no port.
Constraints: parse digits by hand. No atoi/strtol/stoi/sscanf, no inet_*, no regex.
main loops with the prompt "Enter a string (or 'END' to quit): " until the line is exactly "END", then prints "Program terminated." Output formats:
Extracted IPv4 address: A.B.C.D (decimal value: N, port: P|none)
Invalid input: no valid IPv4 address found
Example: "192a168.1.1.1" -> 168.1.1.1, and "192.168.1.1." -> invalid.

*All parts of the code were AI-generated.

Verification statement
I understand all parts of the code and did not blindly copy it. The code has been tested and works according to the assignment guidelines. I did not find any bugs or issues to resolve.